PREDICTING CRIME RATE ANALYSIS IN UNITED STATES

Venkata Gayathri Peri

Sanjeev Peri

Dhirender Kumar Pandey

Vishnu kalyan Chakravarthi Korukonda

CONTEXT

- Introduction
- Understanding of Dataset
- Methodologies/Classifier comparison
- Fitting of Classifier
- Output

INTRODUCTION

The frequency and variety of criminal activity are rising at an alarming rate, pushing organizations to create effective strategies for taking preventive action

predicting crime plays an important role in increasing a country’s economic development .

Datasets and methodologies for predicting crime are many. By using Chicago crime Dataset, we try to mitigate and prevent future crime.

DATASET

- Our dataset shows recorded criminal offenses that took place in the City of Chicago between 2001 to the present.
- Around 10,00,000 observations were jotted down 3,64,000 records to increase the scope for data analysis such that records from 2018 to 2023 are only considered.

ID	Case Num	Date	Block	IUCR	Primary Ty	Descriptio	Location D	Arrest	Domestic	Beat
11646166	JC213529	9/1/2018 0:01	082XX S IN	810	THEFT	OVER $50(	RESIDENC	FALSE	TRUE	631
12014684	JD189901	3/17/2020 21:30	039XX N LI	820	THEFT	$500 AND	STREET	FALSE	FALSE	1634
11645648	JC212959	1/1/2018 8:00	024XX N M	1153	DECEPTIVI	FINANCIAI	RESIDENC	FALSE	FALSE	2515
11864018	JC476123	9/24/2019 8:00	022XX S M	1154	DECEPTIVI	FINANCIAI	COMMER(	FALSE	FALSE	132
11859805	JC471592	10/13/2019 20:30	024XX W (	860	THEFT	RETAIL TH	GROCERY	FALSE	FALSE	1221
12571973	JE482457	12/19/2021 7:23	042XX S M	460	BATTERY	SIMPLE	SIDEWALK	TRUE	TRUE	921
11645959	JC211511	12/20/2018 16:00	045XX N A	2820	OTHER OF	TELEPHON	RESIDENC	FALSE	FALSE	1724
11645557	JC212685	4/1/2018 0:01	080XX S VI	1153	DECEPTIVI	FINANCIAI	RESIDENC	FALSE	FALSE	631

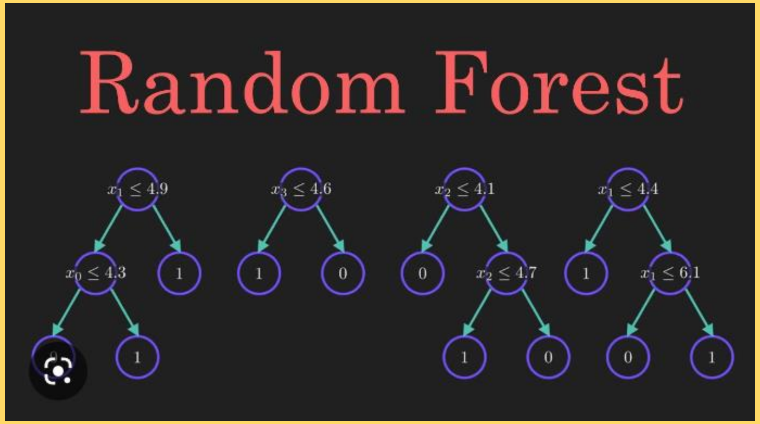
CHICAGO CRIME DATASET

- It contains variables like case number, Description, Location, Arrest and district etc.
- Data is cleaned by removing Null values, missing values from our Dataset.
- We reduced unnecessary columns by using drop method and other pandas' functionalities.
- Data imbalances are handled using resample package to up sample minority class of data

Beat	District	Ward	Communi	FBI Code	X Coordin	Y Coordin	Year	Updated On	Latitude	Longitude	Location
631	6	8	44	6			2018	4/6/2019 16:04			
1634	16	45	15	6	1141659	1925649	2020	3/25/2020 15:45	41.95205	-87.7547	(41.952051946, -87.754660372)
2515	25	30	19	11			2018	4/6/2019 16:04			
132	1	3	33	11	1177560	1889548	2019	10/20/2019 15:56	41.85225	-87.6238	(41.852248185, -87.623786256)
1221	12	26	24	6	1160005	1905256	2019	10/20/2019 16:03	41.89573	-87.6878	(41.895732399, -87.687784384)
921	9	15	58	08B	1158067	1876425	2021	9/12/2022 16:45	41.81666	-87.6957	(41.81665685, -87.695688608)
1724	17	33	14	08A			2018	4/6/2019 16:04			
631	6	6	44	11			2018	4/6/2019 16:04			
2515	25	36	19	11			2018	4/6/2019 16:04			

METHODOLOGIES COMPARISON

K-nearest neighbor (KNN) classifier is a type of instance-based learning algorithm in machine learning and data mining. It is a simple and effective classification algorithm that does not require any training, as it memorizes the entire training dataset. KNN is considered a non-parametric algorithm.



Random Forest is a supervised learning algorithm used for both classification and regression tasks. It is an ensemble learning method that builds multiple decision trees during training and combines their predictions to improve the accuracy of the model.

Accuracy percentage(%)

- KNN -> 33.5%
- RF -> 36.8%

For model selection or methodologies to be chosen, here we compared two algorithms – KNN and Random forest classifier most commonly used for crime data analysis. After calculating the accuracy precision f1 score and recall performance metrics we decided to proceed with random forest classifier to predict our data

Comparison of best models from each category:

Measures	Random Forest	KNN
Accuracy	36.8%	33.5%
F1-Score	25.4%	29.6%

SUMMARY

Run - final.py

Run: g3

```
C:\Users\User\anaconda3\python.exe C:/Users/User/Downloads/g3.py
predictedLabel  Primary Type    Latitude    Longitude   probability0    NARCOTICS   HOMICIDE    41.789497   -87.672963  {BATTERY=0.12389312747887293, HOMICIDE=0.004287664351617304, RO
1   THEFT   HOMICIDE    41.898076   -87.641601  {BATTERY=0.1713619512739252, HOMICIDE=0.0019913028861234215, ROBBERY=0.020810848894596952, SEX OFFENSE=0.02024689092656711, OFFENSE INV
2   NARCOTICS   HOMICIDE    41.770250   -87.674527  {BATTERY=0.1294786228472955, HOMICIDE=0.00426706268133357, ROBBERY=0.024177208012704742, SEX OFFENSE=0.033284870699230834, OFFENSE
3   NARCOTICS   HOMICIDE    41.899876   -87.750819  {BATTERY=0.11123913076564497, HOMICIDE=0.0038606342059982627, ROBBERY=0.02090029492529386, SEX OFFENSE=0.042076607053740896, OFFENS
4   NARCOTICS   HOMICIDE    41.770578   -87.617000  {BATTERY=0.12389312747887293, HOMICIDE=0.004287664351617304, ROBBERY=0.024675434994493085, SEX OFFENSE=0.03402628882875742, OFFENSE

Process finished with exit code 0
```

Run - final.py

Run: g3

```
 SEX OFFENSE=0.03402628882875742, OFFENSE INVOLVING CHILDREN=0.0031874394937005737, NARCOTICS=0.26812021522576096, CRIMINAL SEXUAL ASSAULT=6.565644220574121E-6, CRIMINAL TRESPASS=0.035
51093552, NARCOTICS=0.19602986469404327, CRIMINAL SEXUAL ASSAULT=3.2179483086498435E-5, CRIMINAL TRESPASS=0.08230483161903557, CRIMINAL DAMAGE=0.05265275706085249, DECEPTIVE PRACTICE=
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0.04729037589525074, MOTOR VEHICLE THEFT=0.05456277510700858, BURGLARY=0.006035827375683807, KIDNAPPING=4.041624786199626E-4, THEFT=0.10381947388779822}
.05495839694148831, MOTOR VEHICLE THEFT=0.06105804132567455, BURGLARY=0.007274508463013829, KIDNAPPING=4.4972687407872776E-4, THEFT=0.12795398327097504}
```

OUTPUT

So, after all the data analysis and random forest classifier fitting, we were able to predict the probability for homicide type of crimes in our data.

THANK YOU